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Assignment - 1 (COA)

Section A

Ans 1 Abstraction is the process of hiding complex implementation details and showing only essential features of a system.

we use it because :-
→ Reduces system complexity
→ makes design and development easier
→ Enables modularity

Ans 2) levels of abstraction

1. Hardware Level - Physical component (CPU, memory)
2. Machine level - Binary instructions (0s and 1s)
3. Assembly level - Symbolic instructions (ADD, MOV)
4. High-level language - C, Java, Python
5. Application Level: End-user programs

Example

Print "Hello" → Python code → compiled to assembly → converted to machine code → executed by hardware.

Ans 3) Functional units :-

→ Input unit : Takes input from user
→ Memory unit :- stores data & instructions
→ Arithmetic logic unit → Performs calculations
→ Central unit → controls execution of instructions
→ Output unit → displays result

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Ans4) ALU vs CU

ALU → performs arithmetic and logical operations
CU → controls and coordinates all operations.

Register vs cache

Registers - smallest, fastest storage inside CPU.
Cache - slightly larger, stores frequently used data.

Ans5 Von- neumann Model

A complex architecture where CPU, memory and I/O are connected and instructions/ data share the same memory

Stored-Program concept

instructions are stored in memory like data and executed sequentially.

Ans6 Limitation caused due to a single path (bus) between CPU and memory.

Key points

→ Data and instructions store same bus
→ causes delay in execution
→ limits system performance.

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Ans 7) No. of transistors on a chip doubles approximately every 18-24 months.

Impact :-

→ increased performance
→ Reduced cost
→ smaller and more powerful devices.

Ans 8 IPC (instruction per cycle)

→ IPC is the no. of instruction a CPU executes in one cycle)

→ Higher IPC → Better CPU performance.

Ans 9) Execution Time - Total time to run a program.

Clock Rate - No of cycles per second.

CPI (cycles per instruction) Avg cycle per instruction

IPC - instruction per cycle.

Relation :- Execution Time = (instruction × CPI) / Clock Rate

IPC = 1/CPI

Ans 10 Definitions

Bit - Smallest unit (0 or 1)
Byte - 8 ~~Bytes~~ bits
Signed - Represent positive and negative value
Unsigned - only positive values

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Example

8 bit unsigned :- 0 to 255
8 bit signed :- -128 to +127

Ans 11 Basic operations are :-

- Arithmetic (addition, subtraction, multiplication)
- Logical (AND, OR, NOT)
- Comparision (>, <, =)
- Data movement (Load/Store)

Ans 12) Character Representation

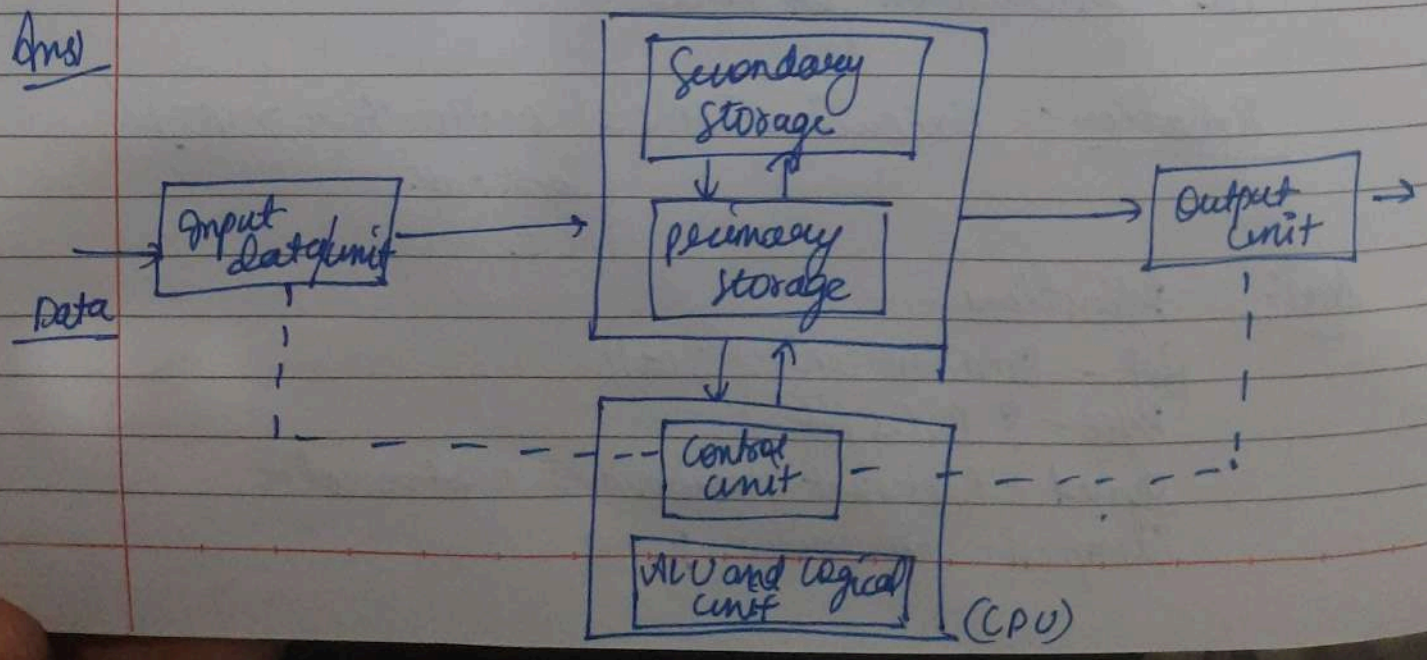
- ASCII (7/8-bit)
- unicode (supports multiple languages)

Decimal Representations

- Fixed point (simple representations)
- Floating point (scientific eg IEEE 754)

Section - B

Ans)



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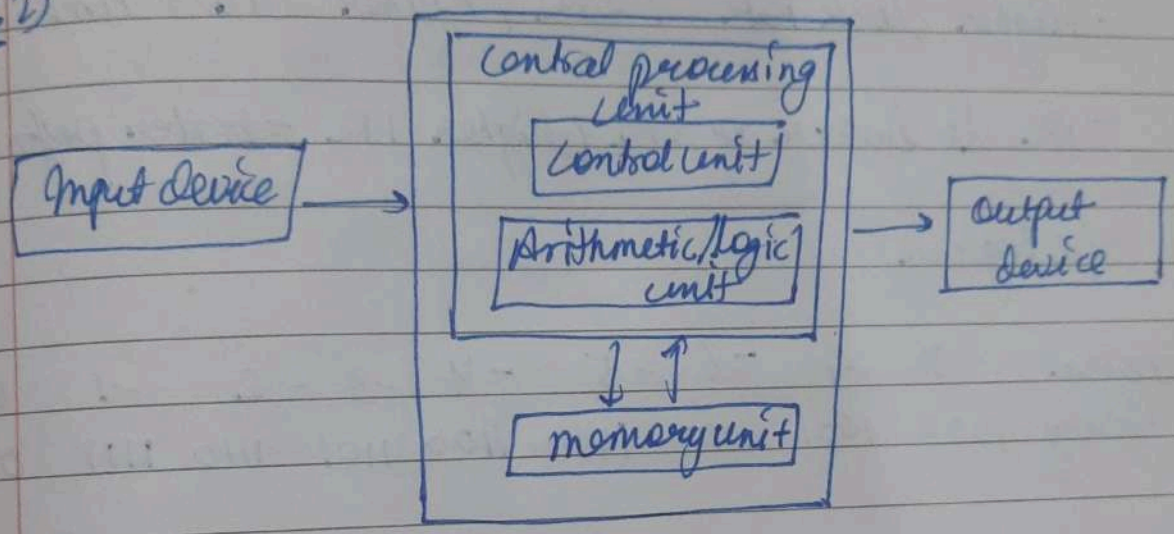
Explanation

- Input unit takes data, CPU processes it using ALU and CU
- memory stores Data and instructions
- Output unit display results : buses connect all unit

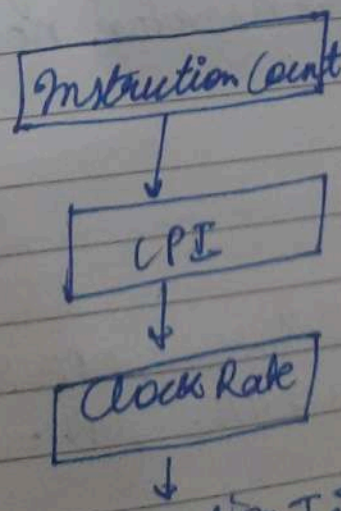
Ans 2)



explanation

- Single memory stores both data & instructions
- CPU communicates with memory and I/o via a common bus
- Basis of most modern computers

Ans 3



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$$\boxed{IPC = 1/CPI}$$

Explanation -

- CPU Time Depends on instruction count, CPI and clock Rate
- Higher clock Rate ↓ time, higher CPI ↑ time
- IPC is inverse of CPI (higher IPC = better performance)

Ans4 (Diagram)

Decimal	-8	-7	-6	-5	-4	-3	-2	-1	0
Binary	1000	1001	1010	1011	1100	1101	1110	1111	0000

1	2	3	4	5	6	7
0000	0001	0010	0010	0011	0100	0101

Explanation

- 4-bit 2's complement range -8 to +7
→ MSB (leftmost bit) indicates sign.
→ used to represent negative nos efficiently.

Section-C

Ans 1) Given

CPU A :- 3.4 Hz, CPI = 1.5
CPU B :- 2.2 GHz, CPI = 1.0

Formula :- CPU Time = (instruction count * CPI)/

Now,

$$CPU\ A = \frac{1.5}{3.0} = 0.5$$

$$CPU\ B\ time = \frac{1.0}{2.2} = \underline{0.45}$$

CPU B has lower execution time, hence faster
Reason :- Lower CPI outweights lower clock rate.

Ans 2 Effect of cache miss

cache miss = CPU waits (stall cycles increases)
- IPC decreases (fewer instructions per cycle)
- execution time increases

fix for it (any one)

- increases cache size
- improved cache locality
- use multi-level cache

Ans 3) Given
Range :- 200 values
Precision :- 0.1 → need decimal accuracy.

Step 1 :- Choice

Used fixed-Point Representation

Step 2 :- Justification
- Fixed-point is sufficient for limited range
- More efficient and faster than floating-point
- less memory usage and simple hardware.

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Ans4 Critique of given statement are mentioned below :-

- Moores Law refers to transistor count, not direct performance.

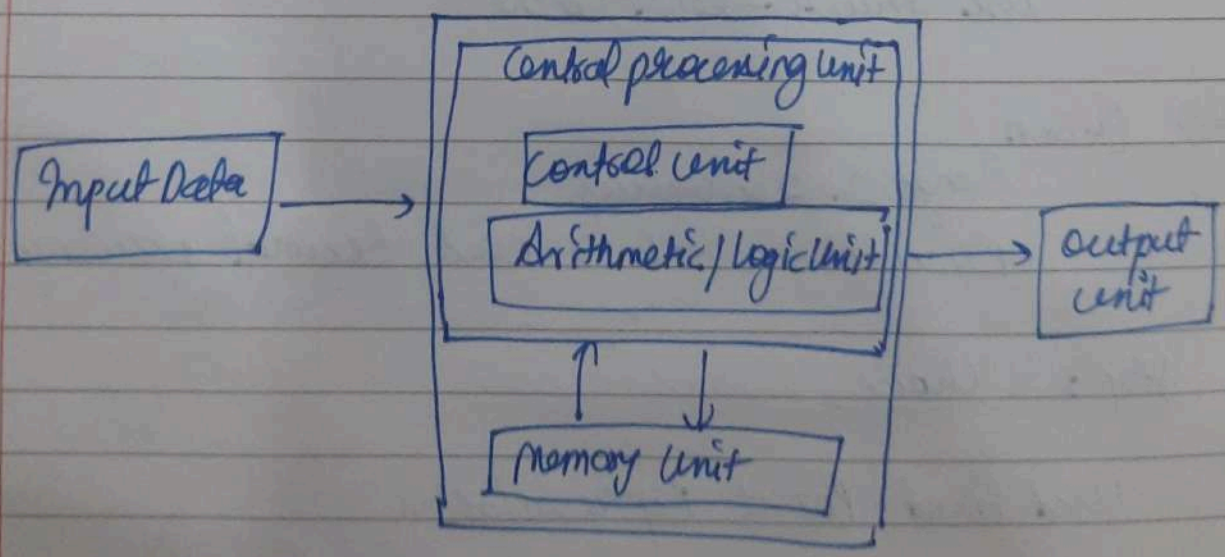
→ Performance gains now limited due to :-
- Power and heat constraints
- slower clock speed scaling.

→ Modern improvement depends on :-
- Parallelism (multicore CPUs)
- Better architecture

Conclusion Moore's Law alone does not guarantee automatic perfance doubling today.

Section-D

Ans1)



Von-Neumann Architecture

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Design Decision :-

- single memory for data and instructions

explanation

- simplifies system design and reduces hardware cost
- But leads to Von Neumann bottleneck due to shared Bus.

Ans 2 Performance Relation

CPU Time = (instruction Count X CPI) / clock Rate

explanation

- performance improves when CPU time decreases
- Depends on IC, CPI and clock Rate

~~Ans~~ Practical Improvement

- Reduce CPI using pipeline or better cache
- Reduce IC using optimized algorithms
- Increases clock rate using better hardware

Ans 3 Represent Number

Let's take +5 and +4 (4 bit system)

+5 → 0101
+4 → 0100

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Addition

$$\begin{array}{r} 0101 \\ +\,0100 \\ \hline 1001 \\ \hline \end{array}$$

interpretation

1001 in 2's complement = −7

Overflow check

- adding two positive nos → result should be positive
- But result is negative → overflow occurs.

→ In conclusion, overflow happens when result exceeds representation range.